MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 1: Fundamentals and Particle Kinematics - Problem Sets

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Ground Truth: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

Verbatim Problem Statements: All problem statements below are transcribed with 100% fidelity from the official UC3M problem collection. Solutions will be added in subsequent analytical development phases.

PROBLEM 01

Rocket Helical Trajectory

Original statement · Problems.pdf, p. 6

MIT OCW Benchmark

An out-of-control rocket traces an upward helical trajectory described in Cartesian coordinates for $t > 0$ by:

$$x = a\cos(\omega t) \tag{1}$$

$$y = a\sin(\omega t) \tag{2}$$

$$z = v_z t \tag{3}$$

where a, v_z, and ω are constants.

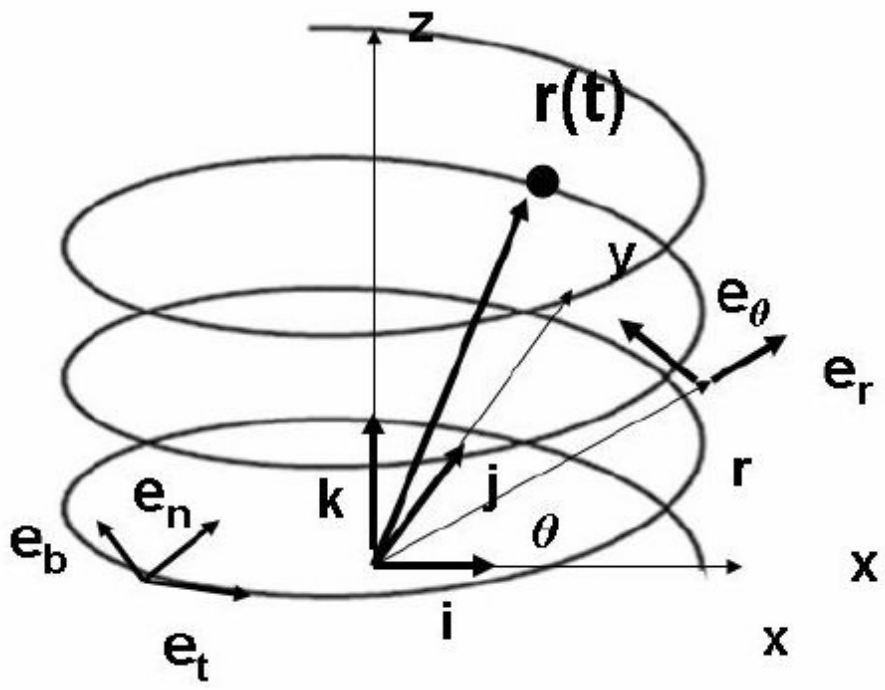


Figure: Helical trajectory with position r(t), the Cartesian basis i, j, k, the cylindrical basis e_R, e_theta, e_z and the intrinsic basis e_t, e_n, e_b (taken from MIT OCW) (Problems.pdf, p. 6).

(a) Express the position, velocity, and acceleration vectors in the Cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

(b) Express the position, velocity, and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.

(c) Express the velocity and acceleration vectors using s, ρ and the vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.

(d) Compute the modulus of the velocity vector.

(e) Express the evolution of the arc-length coordinate as a function of time.

(f) Write down the trajectory as a function of the arc-length coordinate.

(g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** an out-of-control rocket, modelled as a point particle P, ascends along a circular helix around the vertical z-axis.
- **Hypotheses:** $a > 0$, $v_z > 0$ and $\omega > 0$ are constants and $t > 0$.
- **Constraints and degrees of freedom:** the statement gives the three Cartesian coordinates as functions of time, so the particle is on the helix, which satisfies the two holonomic relations $x = a\cos(\omega z/v_z)$ and $y = a\sin(\omega z/v_z)$ (substituting $z = v_z t$ recovers the statement). Two independent relations leave $\mathrm{CDOF} = 3 - 2 = 1$: a single coordinate ($t$, or the arc length s) fixes the position on the helix.
- **Coordinates:** the Cartesian coordinates (x, y, z) of the statement; for the questions we also use the cylindrical coordinates (R, θ, z) and the arc length s.
- **Parameters:** radius a [m], angular frequency ω [rad/s], vertical velocity v_z [m/s], time t [s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, fixed to the launch site with $\mathbf{k}_0$ along the vertical z-axis.
- **Cylindrical basis:** $\mathcal{B}_{\mathrm{cyl}} = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$, a rotation of $\mathcal{B}_0$ by the angle $\theta = \omega t$ about $\mathbf{k}_0$:

$$[{}_0R_{\mathrm{cyl}}] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_{\mathrm{cyl}}] = \cos^2(\omega t) + \sin^2(\omega t) = 1, \qquad [{}_0R_{\mathrm{cyl}}][{}_0R_{\mathrm{cyl}}]^T = I$$

- The basis vectors, from the columns of the matrix:

$$\mathbf{e}_R = \cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0, \qquad \mathbf{e}_\theta = -\sin(\omega t)\mathbf{i}_0 + \cos(\omega t)\mathbf{j}_0, \qquad \mathbf{e}_z = \mathbf{k}_0$$

- **Time derivatives of the unit vectors** (direct differentiation of their explicit components, with the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$ and $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$; Notes Sec. 2.6.2):

$$\frac{d\mathbf{e}_R}{dt} = -\omega\sin(\omega t)\,\mathbf{i}_0 + \omega\cos(\omega t)\,\mathbf{j}_0 = \omega\,\mathbf{e}_\theta, \qquad \frac{d\mathbf{e}_\theta}{dt} = -\omega\cos(\omega t)\,\mathbf{i}_0 - \omega\sin(\omega t)\,\mathbf{j}_0 = -\omega\,\mathbf{e}_R, \qquad \frac{d\mathbf{e}_z}{dt} = \mathbf{0}$$

- **Intrinsic basis:** $\mathcal{B}_F = \{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ (tangent, principal normal toward the centre of curvature, binormal $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$), obtained in 3.1 and related to $\mathcal{B}_{\text{cyl}}$ by a second change-of-basis matrix.

PHASE 3

Derivation

3.1 Kinematics

(a) Cartesian position, velocity and acceleration

Pedagogical Justification: the position is given by its Cartesian components in the fixed basis, whose unit vectors are constant ($\dot{\mathbf{i}}_0 = \dot{\mathbf{j}}_0 = \dot{\mathbf{k}}_0 = \mathbf{0}$). Velocity and acceleration follow by differentiating term by term.

$$\mathbf{r}(t) = a\cos(\omega t)\,\mathbf{i}_0 + a\sin(\omega t)\,\mathbf{j}_0 + v_z t\,\mathbf{k}_0$$

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = -a\omega\sin(\omega t)\,\mathbf{i}_0 + a\omega\cos(\omega t)\,\mathbf{j}_0 + v_z\,\mathbf{k}_0$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = -a\omega^2\cos(\omega t)\,\mathbf{i}_0 - a\omega^2\sin(\omega t)\,\mathbf{j}_0 + 0\,\mathbf{k}_0 = -a\omega^2\left(\cos(\omega t)\,\mathbf{i}_0 + \sin(\omega t)\,\mathbf{j}_0\right)$$

(b) Cylindrical position, velocity and acceleration

> **Pedagogical Justification:** the helix lies on a cylinder, so $R = \sqrt{x^2+y^2} = a$ is constant and $\theta = \omega t$. The vectors $\mathbf{e}_R$ and $\mathbf{e}_\theta$ are not constant, so we use the product rule with the derivatives of Phase 2.

$$\mathbf{r} = a\,\mathbf{e}_R + v_z t\,\mathbf{e}_z$$

$$\mathbf{v} = \dot{a}\,\mathbf{e}_R + a\frac{d\mathbf{e}_R}{dt} + \frac{d(v_z t)}{dt}\mathbf{e}_z + v_z t\frac{d\mathbf{e}_z}{dt} = 0 + a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z + 0 = a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z$$

$$\mathbf{a} = \frac{d(a\omega)}{dt}\mathbf{e}_\theta + a\omega\frac{d\mathbf{e}_\theta}{dt} + \frac{dv_z}{dt}\mathbf{e}_z + v_z\frac{d\mathbf{e}_z}{dt} = 0 + a\omega\left(-\omega\,\mathbf{e}_R\right) + 0 + 0 = -a\omega^2\,\mathbf{e}_R$$

(d) Modulus of the velocity

> **Pedagogical Justification:** the speed is the norm of the velocity vector, the rate at which the distance along the path is travelled. It is computed before part (c) because the intrinsic basis and the radius of curvature need it.

$$v = \|\mathbf{v}\| = \sqrt{a^2\omega^2\left(\sin^2\omega t + \cos^2\omega t\right) + v_z^2} = \sqrt{a^2\omega^2 + v_z^2} \equiv V$$

The speed is constant for all t.

(c) Intrinsic basis, and velocity and acceleration in terms of s and ρ

> **Pedagogical Justification:** the Frenet-Serret basis follows the curve itself (Notes Sec. 2.4): $\mathbf{e}_t$ is the direction of the velocity, $\mathbf{e}_n$ points toward the centre of the osculating circle and $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ completes a right-handed triad. The intrinsic acceleration is (Notes Eq. 2.17)

$$\mathbf{a} = \ddot{s}\,\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\,\mathbf{e}_n, \qquad \mathbf{v} = \dot{s}\,\mathbf{e}_t$$

- **Tangent.** $\mathbf{e}_t = \dfrac{\mathbf{v}}{\|\mathbf{v}\|} = \dfrac{a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z}{V}$.
- **Principal normal.** Since $\dot{s} = V$ is constant, $\ddot{s} = dV/dt = 0$ and the whole acceleration is normal: $\mathbf{a} = a_n \mathbf{e}_n = -a\omega^2 \mathbf{e}_R$. Taking the norm gives $a_n = a\omega^2$, and dividing the vector by it gives

$$\mathbf{e}_n = \frac{\mathbf{a}}{a_n} = \frac{-a\omega^2 \mathbf{e}_R}{a\omega^2} = -\mathbf{e}_R$$

The principal normal points radially inward, toward the axis of the cylinder.

- **Radius of curvature.** From $a_n = \dot{s}^2/\rho$:

$$\rho = \frac{V^2}{a_n} = \frac{a^2\omega^2 + v_z^2}{a\omega^2} = a + \frac{v_z^2}{a\omega^2}$$

- **Binormal.** Using $\mathbf{e}_R \times \mathbf{e}_\theta = \mathbf{e}_z$ and its cyclic permutations ($\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\theta$), together with the anticommutativity $\mathbf{e}_\theta \times \mathbf{e}_R = -\mathbf{e}_z$:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \frac{a\omega\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z}{V} \times (-\mathbf{e}_R) = \frac{-a\omega\,(\mathbf{e}_\theta \times \mathbf{e}_R) - v_z\,(\mathbf{e}_z \times \mathbf{e}_R)}{V} = \frac{-v_z\,\mathbf{e}_\theta + a\omega\,\mathbf{e}_z}{V}$$

- **Velocity and acceleration with s and ρ:**

$$\mathbf{v} = \dot{s}\,\mathbf{e}_t = V\,\mathbf{e}_t, \qquad \mathbf{a} = \ddot{s}\,\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\,\mathbf{e}_n = \frac{V^2}{\rho}\,\mathbf{e}_n = a\omega^2\,\mathbf{e}_n$$

- **Second change-of-basis matrix.** Its columns are the components of $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_{\text{cyl}}$:

$$[{}_{\text{cyl}}R_{\text{tnb}}] = \begin{pmatrix} 0 & -1 & 0 \\ a\omega/V & 0 & -v_z/V \\ v_z/V & 0 & a\omega/V \end{pmatrix}, \qquad RR^T = I$$

Expanding the determinant along the first row (its only non-zero entry is -1 in position $(1,2)$, with cofactor sign -1):

$$\det = -(-1)\left[\frac{a\omega}{V}\cdot\frac{a\omega}{V} - \left(-\frac{v_z}{V}\right)\frac{v_z}{V}\right] = \frac{a^2\omega^2 + v_z^2}{V^2} = 1$$

The columns have unit norm and are mutually orthogonal, so $RR^T = I$; the composition $[{}_0R_{\text{tnb}}] = [{}_0R_{\text{cyl}}][{}_{\text{cyl}}R_{\text{tnb}}]$ is again a proper rotation (determinant $1 \cdot 1 = 1$).

3.2 Forces

Not applicable: kinematics-only problem. The trajectory is given, so no force enters the question.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by the statement, so there is no equation of motion to solve.

3.4 Integration and Answers

(e) Arc length as a function of time

> **Pedagogical Justification:** $s(t)$ measures the distance travelled along the trajectory from $t = 0$, i.e. the integral of the speed.

$$s(t) = \int_0^t v\,dt' = \int_0^t V\,dt' = V\,[t']_0^t = V(t-0) = t\sqrt{a^2\omega^2 + v_z^2}$$

(f) Trajectory as a function of the arc length

> **Pedagogical Justification:** inverting $s(t)$ gives $t(s) = s/V$, which is substituted into the parametric equations to remove the time:

$$x(s) = a\cos\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right), \qquad y(s) = a\sin\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right), \qquad z(s) = \frac{v_z s}{\sqrt{a^2\omega^2 + v_z^2}}$$

(g) Centre of curvature at $(a, 0, 0)$

> **Pedagogical Justification:** the centre C of the osculating circle lies on the principal normal at distance ρ: $\mathbf{r}_C = \mathbf{r}_P + \rho\,\mathbf{e}_n$ (Notes Sec. 2.4).

- The helix passes through $(a, 0, 0)$ where $z = v_z t = 0$, i.e. at $t = 0$. The statement says $t > 0$, so we take the motion law to hold for $t \geq 0$ (the point is the limit $t \to 0^+$; all formulas are continuous there). At that point $\mathbf{r}_P(0) = a\,\mathbf{i}_0$ and $\mathbf{e}_n(0) = -\mathbf{e}_R(0) = -\mathbf{i}_0$:

$$\mathbf{r}_C = a\,\mathbf{i}_0 + \left(a + \frac{v_z^2}{a\omega^2}\right)(-\mathbf{i}_0) = -\frac{v_z^2}{a\omega^2}\,\mathbf{i}_0, \qquad \boxed{C = \left(-\frac{v_z^2}{a\omega^2},\ 0,\ 0\right)}$$

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[a\omega] = \mathrm{m} \cdot \mathrm{s}^{-1}$ and $[v_z] = \mathrm{m/s}$, so V is in m/s. The offset $\left[\dfrac{v_z^2}{a\omega^2}\right] = \dfrac{(\mathrm{m/s})^2}{\mathrm{m} \cdot \mathrm{s}^{-2}} = \mathrm{m}$.
- **Limiting cases:** if $v_z \to 0$ (planar circle) $\rho = a$ and $C = (0, 0, 0)$, the centre of the circle. If $\omega \to 0$ (vertical ascent) $\rho \to \infty$: the curvature vanishes and the trajectory is a straight line.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 1 in its solution keys. The results are verified by independent derivation: sympy reproduces $\|\mathbf{v}\| = V$, $a_n = a\omega^2$, $\rho = a + v_z^2/(a\omega^2)$ (also from $\rho = \|\mathbf{v}\|^3 / \|\mathbf{v} \times \mathbf{a}\|$), the binormal, the centre of curvature and $\det[{}_{\mathrm{cyl}}R_{\mathrm{tnb}}] = 1$ with $RR^T = I$.

Source: Problems.pdf (Taken from MIT OCW)

PROBLEM 02

Kinematics, Spiral

Original statement · Problems.pdf, p. 6

Polar Kinematics

A planar spiral is given by the equation $r = \alpha e^{\beta\theta}$, where α and β are constants, r is the radial coordinate, and θ is the angular coordinate. If a particle moves along the spiral according to the equation $\theta = \delta t^n$ (where δ is a constant), determine:

(a) The radial and transverse components of the velocity.

(b) The radial and transverse components of the acceleration.

(c) The tangential and normal components of the velocity.

(d) The tangential and normal components of the acceleration.

(e) The radius of curvature of the spiral.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P moves along the planar spiral $r = \alpha e^{\beta\theta}$ with the prescribed angular motion $\theta(t) = \delta t^n$.
- **Hypotheses:** $\alpha, \beta, \delta, n > 0$ are constants and $t > 0$. At $t = 0$ the speed vanishes for $n > 1$ and $\dot\theta$ diverges for $n < 1$, and $\ddot\theta \propto t^{n-2}$ diverges for $n < 2$ except at $n = 1$; only $n = 1$ is regular at $t = 0$, and $t > 0$ is kept for all n for uniformity. The motion is planar ($z = 0$).

- **Constraints and degrees of freedom:** the spiral $r = \alpha e^{\beta\theta}$ is a holonomic, scleronomic constraint in the plane. The particle then has $\mathrm{CDOF} = 2 - 1 = 1$, and the motion law $\theta = \delta t^n$ fixes it completely.
- **Coordinates:** polar coordinates (r, θ) about the origin O of the spiral; the angle θ is the free coordinate.
- **Parameters and units:** $[r] = \mathrm{m}$, so $[\alpha] = \mathrm{m}$; β is dimensionless; $[\theta] = \mathrm{rad}$, so $[\delta] = \mathrm{rad} \cdot \mathrm{s}^{-n}$.

PHASE 2

Reference Frames & Vector Bases

- **Inertial basis:** $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Polar basis:** $\mathcal{B}_{\mathrm{polar}} = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$, a rotation of $\mathcal{B}_0$ by θ about $\mathbf{k}_0$:

$$[{}_0R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_\theta] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_\theta][{}_0R_\theta]^T = I$$

- **Angular velocity of the basis:** $\boldsymbol{\omega} = \dot\theta\, \mathbf{k}_0$. By Poisson's formula (Notes Sec. 2.6.2, Eqs. 2.25-2.26): $\dot{\mathbf{e}}_r = \dot\theta\, \mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\, \mathbf{e}_r$.
- **Intrinsic basis from the polar basis** (derived in 3.1(c), where the sign of the normal is justified): $\mathbf{e}_t = (\beta\, \mathbf{e}_r + \mathbf{e}_\theta)/\sqrt{1+\beta^2}$, $\mathbf{e}_n = (-\mathbf{e}_r + \beta\, \mathbf{e}_\theta)/\sqrt{1+\beta^2}$ and $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \mathbf{k}_0$. The components of $\mathbf{e}_t$ and $\mathbf{e}_n$ in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are the columns of

$$[{}_{\mathrm{polar}}R_{\mathrm{tn}}] = \frac{1}{\sqrt{1+\beta^2}} \begin{pmatrix} \beta & -1 \\ 1 & \beta \end{pmatrix}, \qquad \det = \frac{\beta^2+1}{1+\beta^2} = 1, \qquad RR^T = I$$

This is a planar rotation by the constant angle $\arctan(1/\beta)$: the tangent makes a constant angle with the radial direction (equiangular spiral).

PHASE 3

Derivation

3.1 Kinematics

(a) Radial and transverse components of the velocity

Pedagogical Justification: in polar coordinates $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta$ (product rule and Poisson). The coordinates depend on time through $\theta(t)$, so the chain rule is needed.

$$\dot{\theta} = \frac{d}{dt}\left(\delta t^n\right) = n\delta t^{n-1}, \qquad \frac{dr}{d\theta} = \frac{d}{d\theta}\left(\alpha e^{\beta\theta}\right) = \alpha\beta e^{\beta\theta} = \beta r$$

$$\dot{r} = \frac{dr}{d\theta}\,\dot{\theta} = \beta r\, n\delta t^{n-1} = \alpha\beta n\delta t^{n-1} e^{\beta\delta t^n}$$

$$\boxed{v_r = \dot{r} = \beta n\delta t^{n-1}\, r}, \qquad \boxed{v_\theta = r\dot{\theta} = n\delta t^{n-1}\, r}$$

(b) Radial and transverse components of the acceleration

Pedagogical Justification: the polar acceleration is (Notes Eq. 2.30; product rule applied to $\mathbf{v}$ with Poisson's formula)

$$\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

- **Second derivatives** (product rule, with $\dot{r} = \beta r\dot{\theta}$):

$$\ddot{\theta} = \frac{d}{dt}\left(n\delta t^{n-1}\right) = n(n-1)\delta t^{n-2}, \qquad \ddot{r} = \beta\left(\dot{r}\dot{\theta} + r\ddot{\theta}\right) = \beta\left(\beta r\dot{\theta}^2 + r\ddot{\theta}\right) = \beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}$$

- **Radial component:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = \left(\beta^2 - 1\right) r\dot{\theta}^2 + \beta r\ddot{\theta} = n\delta t^{n-2}\, r\left[\left(\beta^2 - 1\right) n\delta t^n + \beta(n-1)\right]$$

- **Transverse component:**

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\ddot{\theta} + 2\beta r\dot{\theta}^2 = n\delta t^{n-2}\, r\left[(n-1) + 2\beta n\delta t^n\right]$$

(c) Tangential and normal components of the velocity

> **Pedagogical Justification:** the velocity is always tangent to the path, so in the intrinsic basis $\mathbf{v} = v\,\mathbf{e}_t + 0\,\mathbf{e}_n$ and the normal component is identically zero.

$$v_t = v = \sqrt{v_r^2 + v_\theta^2} = r\dot{\theta}\sqrt{\beta^2 + 1} = n\delta t^{n-1}\, r\sqrt{1 + \beta^2}, \qquad v_n = 0$$

- The unit tangent follows from $\mathbf{v} = r\dot{\theta}\,(\beta\,\mathbf{e}_r + \mathbf{e}_\theta)$: $\mathbf{e}_t = (\beta\,\mathbf{e}_r + \mathbf{e}_\theta)\,/\sqrt{1+\beta^2}$. The normal toward the centre of curvature must satisfy $\mathbf{a}\cdot\mathbf{e}_n > 0$. The left normal $\mathbf{k}_0 \times \mathbf{e}_t = (-\mathbf{e}_r + \beta\,\mathbf{e}_\theta)\,/\sqrt{1+\beta^2}$ does: its projection of the acceleration (with a_r and a_θ of part (b)) is $(\beta a_\theta - a_r)\,/\sqrt{1+\beta^2}$, and

$$\beta a_\theta - a_r = \beta\left(r\ddot{\theta} + 2\beta r\dot{\theta}^2\right) - \left(\left(\beta^2 - 1\right) r\dot{\theta}^2 + \beta r\ddot{\theta}\right) = \left(1 + \beta^2\right) r\dot{\theta}^2 > 0$$

so $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t$ and $\mathbf{e}_b = \mathbf{k}_0$.

(d) Tangential and normal components of the acceleration

> **Pedagogical Justification:** $a_t = \dot{v}$ is the rate of change of the speed and $a_n = v^2/\rho$ measures the turning of the velocity.

$$a_t = \frac{d}{dt}\left(\sqrt{1+\beta^2}\, r\dot{\theta}\right) = \sqrt{1+\beta^2}\left(\dot{r}\dot{\theta} + r\ddot{\theta}\right) = \sqrt{1+\beta^2}\left(\beta r\dot{\theta}^2 + r\ddot{\theta}\right) = \sqrt{1+\beta^2}\, n\delta t^{n-2}\, r\left[\beta n\delta t^n + (n-1)\right]$$

- **Normal component.** The cross product $\mathbf{a}\times\mathbf{v} = (a_t\mathbf{e}_t + a_n\mathbf{e}_n)\times v\,\mathbf{e}_t = -v a_n\,\mathbf{e}_b$ gives $a_n = \dfrac{\|\mathbf{v}\times\mathbf{a}\|}{v} = \dfrac{|v_r a_\theta - v_\theta a_r|}{v}$. The numerator is

$$v_r a_\theta - v_\theta a_r = \left(\beta r\dot{\theta}\right)\left(r\ddot{\theta} + 2\beta r\dot{\theta}^2\right) - \left(r\dot{\theta}\right)\left(\left(\beta^2-1\right) r\dot{\theta}^2 + \beta r\ddot{\theta}\right)$$

$$= \beta r^2\dot{\theta}\ddot{\theta} + 2\beta^2 r^2\dot{\theta}^3 - \left(\beta^2-1\right) r^2\dot{\theta}^3 - \beta r^2\dot{\theta}\ddot{\theta} = \left(\beta^2+1\right) r^2\dot{\theta}^3$$

Dividing by $v = r\dot{\theta}\sqrt{\beta^2+1}$:

$$a_n = \frac{\left(\beta^2+1\right) r^2\dot{\theta}^3}{r\dot{\theta}\sqrt{\beta^2+1}} = r\dot{\theta}^2\sqrt{\beta^2+1} = n^2\delta^2 t^{2n-2}\, r\sqrt{1+\beta^2}$$

3.2 Forces

Not applicable: kinematics-only problem. The path and the motion law are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by $\theta(t) = \delta t^n$.

3.4 Integration and Answers

(e) Radius of curvature

> **Pedagogical Justification:** the radius of curvature follows from the normal acceleration, $a_n = v^2/\rho$:

$$\rho = \frac{v^2}{a_n} = \frac{r^2\dot{\theta}^2\left(1+\beta^2\right)}{r\dot{\theta}^2\sqrt{1+\beta^2}} = \boxed{r\sqrt{1+\beta^2}}$$

PHASE 4

Interpretation & Checks

- **Geometric invariance:** on the spiral the angle ψ between the radial line and the tangent satisfies $\tan\psi = r/(dr/d\theta) = 1/\beta$, a constant. The spiral is self-similar: $\lambda\, r(\theta) = r(\theta + \ln\lambda/\beta)$, so scaling it by a factor λ gives the same curve rotated by an angle of magnitude $\ln\lambda/\beta$. Hence every length of the curve, ρ included, scales with r.
- **Dimensional check:** $[\rho] = [r]\left[\sqrt{1+\beta^2}\right] = \mathrm{m}$; $[v_r] = [v_\theta] = [r\dot{\theta}] = \mathrm{m/s}$, and $[a_n] = [r\dot{\theta}^2] = \mathrm{m/s^2}$.
- **Limiting case $\beta \to 0$:** the spiral degenerates into the circle $r = \alpha$, with $\rho = r$ and $a_r = -r\dot{\theta}^2$, the centripetal acceleration.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 2 in its solution keys. The results are verified by independent derivation: sympy reproduces $v_r/(r\dot{\theta}) = \beta$, $v_\theta/(r\dot{\theta}) = 1$, the factorised forms of a_r, a_θ and a_t, the identity $a_n^2 = r^2\dot{\theta}^4\left(1+\beta^2\right)$ and $\rho = r\sqrt{1+\beta^2}$, with zero residual.

Source: Problems.pdf

PROBLEM 03

Kinematics, Astroid

Original statement · Problems.pdf, p. 6

Intrinsic Kinematics

A point particle describes the astroid:

$$x = a\cos^3\phi, \qquad y = a\sin^3\phi$$

with a motion whose tangential acceleration is constant, k. If the point is initially at rest at $\phi = 0$, compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. *(Hint:* $ds = \sqrt{dx^2 + dy^2}$*).*

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P moves on the astroid $x = a\cos^3\phi, y = a\sin^3\phi$ with constant tangential acceleration $a_t = k$ (the statement calls it k) and starts from rest at $\phi = 0$.
- **Hypotheses:** $a > 0, k > 0$. The motion stays within one smooth segment of the curve, here the first-quadrant arc $0 \le \phi \le \pi/2$ (the astroid has cusps at $\phi = 0, \pi/2, \pi, 3\pi/2$). Starting at the cusp $\phi = 0$ the particle could enter either neighbouring arc; we choose the one with ϕ increasing, so that s increases with ϕ and $d\phi \ge 0$.
- **Constraints and degrees of freedom:** the curve is a holonomic, scleronomic constraint in the plane, so $\mathrm{CDOF} = 2 - 1 = 1$. The coordinate is ϕ, or the arc length s measured from $\phi = 0$.
- **Parameters and units:** scale a [m], tangential acceleration k [m/s^2], time t [s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with the curve in the Oxy plane.
- **Parametric curve:** $x(\phi) = a\cos^3\phi$, $y(\phi) = a\sin^3\phi$. Differentiating with respect to the parameter (chain rule):

$$\frac{dx}{d\phi} = 3a\cos^2\phi\,(-\sin\phi) = -3a\cos^2\phi\sin\phi, \qquad \frac{dy}{d\phi} = 3a\sin^2\phi\cos\phi$$

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: we use the arc length as the coordinate along the path. The differential arc length follows from the hint, $ds = \sqrt{dx^2 + dy^2}$:

$$ds^2 = \left(-3a\cos^2\phi\sin\phi\right)^2 d\phi^2 + \left(3a\sin^2\phi\cos\phi\right)^2 d\phi^2 = 9a^2\cos^4\phi\sin^2\phi\,d\phi^2 + 9a^2\sin^4\phi\cos^2\phi\,d\phi^2$$

$$ds^2 = 9a^2\cos^2\phi\sin^2\phi\left(\cos^2\phi + \sin^2\phi\right)d\phi^2 = 9a^2\cos^2\phi\sin^2\phi\,d\phi^2$$

For $0 \le \phi \le \pi/2$ both $\sin\phi$ and $\cos\phi$ are non-negative (and $d\phi \ge 0$), so the positive root is

$$ds = 3a\sin\phi\cos\phi\,d\phi$$

- **Tangential acceleration.** The tangential component of the acceleration is the second derivative of the arc length: $a_t = \ddot{s} = k$.

3.2 Forces

Not applicable: kinematics-only problem. The tangential acceleration is given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The law $\ddot{s} = k$ is the statement itself.

3.4 Integration and Answers

Step 1: arc length as a function of time

> **Pedagogical Justification:** integrate $\ddot{s} = k$ twice, with the initial conditions $\dot{s}(0) = 0$ (at rest) and $s(0) = 0$ (at $\phi = 0$):

$$\dot{s}(t) = \int_0^t k\,dt' = [kt']_0^t = kt - 0 = kt, \qquad s(t) = \int_0^t kt'\,dt' = \left[\frac{1}{2}kt'^2\right]_0^t = \frac{1}{2}kt^2 - 0 = \frac{1}{2}kt^2$$

Step 2: arc length as a function of ϕ

> **Pedagogical Justification:** integrate ds from $\phi = 0$. With the substitution $w = \sin u$, $dw = \cos u\,du$, the limits become $w = 0$ at $u = 0$ and $w = \sin\phi$ at $u = \phi$:

$$s(\phi) = \int_0^{\phi} 3a \sin u \cos u\,du = 3a \int_0^{\sin\phi} w\,dw = 3a\left[\frac{w^2}{2}\right]_0^{\sin\phi} = 3a\left(\frac{\sin^2\phi}{2} - 0\right) = \frac{3a}{2}\sin^2\phi$$

Step 3: equating both expressions

> **Pedagogical Justification:** the distance travelled along the curve is the same quantity in both descriptions, $s(t) = s(\phi)$:

$$\frac{3a}{2}\sin^2\phi = \frac{1}{2}kt^2 \implies \sin^2\phi = \frac{kt^2}{3a} \implies \sin\phi = \sqrt{\frac{k}{3a}}\,t$$

where the positive root is taken because $0 \le \phi \le \pi/2$. Therefore

$$\boxed{\phi(t) = \arcsin\left(\sqrt{\frac{k}{3a}}\, t\right)}$$

Step 4: position as a function of time

Pedagogical Justification: with $\sin\phi = \sqrt{k/(3a)}\, t$ and $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1 - kt^2/(3a)}$ (non-negative in the first quadrant):

$$\boxed{x(t) = a\cos^3\phi = a\left(1 - \frac{kt^2}{3a}\right)^{3/2}}, \qquad \boxed{y(t) = a\sin^3\phi = a\left(\frac{k}{3a}\right)^{3/2} t^3}$$

These hold for $0 \le t \le t_{\text{cusp}}$, where t_{cusp} is defined in Phase 4.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $\left[\sqrt{\frac{k}{3a}}\, t\right] = \sqrt{\frac{\mathrm{m\,s^{-2}}}{\mathrm{m}}}\,\mathrm{s} = \mathrm{s^{-1}}\cdot\mathrm{s} = 1$, so the argument of arcsin is dimensionless, as it must be.
- **Physical limits:** at $t = 0$, $\phi = 0$ and the particle is at the cusp $(a, 0)$. It reaches the next cusp, $\phi = \pi/2$ at $(0, a)$, when $\sin\phi = 1$:

$$\sqrt{\frac{k}{3a}}\, t_{\text{cusp}} = 1 \implies t_{\text{cusp}} = \sqrt{\frac{3a}{k}}$$

The total length of the arc is $s(\pi/2) = 3a/2$, consistent with $\frac{1}{2}kt_{\text{cusp}}^2 = \frac{1}{2}k\cdot\frac{3a}{k} = \frac{3a}{2}$. The solution stops at the cusp for a physical reason. The tangent is $\mathbf{e}_t = (-\cos\phi, \sin\phi)$ and $d\mathbf{e}_t/d\phi = (\sin\phi, \cos\phi)$ has unit norm, so $\rho = \dfrac{ds/d\phi}{\|d\mathbf{e}_t/d\phi\|} = 3a\sin\phi\cos\phi$, which tends to 0 at both cusps, with the centre of curvature on the normal $\mathbf{e}_n = (\sin\phi, \cos\phi)$ (the arc is convex toward the origin). At arrival the speed is $\dot{s}(t_{\text{cusp}}) = kt_{\text{cusp}} = \sqrt{3ak} \neq 0$, so $a_n =$

$\dot{s}^2/\rho \to \infty$: the motion cannot continue with constant a_t through the cusp. At the start, $a_n \approx \dfrac{(kt)^2}{3a\phi} \to 0$ because $\phi \approx \sqrt{k/(3a)}\,t$, so the departure cusp is regular. Explicitly, $a_n \approx \dfrac{k^2t^2}{3a\sqrt{k/(3a)}\,t} = k\sqrt{\dfrac{k}{3a}}\,t \to 0$.

- **Check against the official key:** the key of `Problems.pdf` is $\phi = \arcsin\left(\sqrt{k/(3a)}\,t\right)$ and $s = kt^2/2$ (read on the rendered page), which equals the result above. Sympy also confirms that $s(\phi(t)) - kt^2/2 = 0$ identically.

Source: Problems.pdf

PROBLEM 04

Plane at Constant Altitude

Original statement · Problems.pdf, p. 7

Coordinate Transformations

An aircraft moves at constant altitude y_0 with velocity v_0 in the x direction. At time $t = 0$ it is located at $x = 0$, and it begins to accelerate in the x direction at a constant rate a_x. Describe the position, velocity, and acceleration in polar coordinates r, θ, and answer the following questions:

(a) Show that: $\dot{r} = \frac{v(t)x(t)}{r(t)}$

(b) Show that: $\dot{\theta} = -\frac{v(t)y_0}{r^2(t)}$

(c) Show that: $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r^3(t)}$

(d) Show that: $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r^4(t)}$

From these expressions form the acceleration vector $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$. Repeat the problem using Cartesian coordinates.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** an aircraft moves at constant altitude y_0 with velocity v_0 along x. At $t = 0$ it is at $x = 0$ and begins to accelerate along x at a constant rate a_x. A fixed observer at the origin O describes its position with polar coordinates.
- **Hypotheses:** $y_0 > 0$, $v_0 \geq 0$ and a_x are constants.

- **Constraints and degrees of freedom:** the altitude is fixed, $y = y_0$ (holonomic, scleronomic, so $\dot{y} = \ddot{y} = 0$). The aircraft has $\mathrm{CDOF} = 2 - 1 = 1$, with the coordinate $x(t)$ given by the statement.
- **Coordinates:** Cartesian $x(t)$ for the motion, and polar (r, θ) about O for the description requested, with θ measured from $+\mathbf{i}$.

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and $\mathbf{r} = x\,\mathbf{i} + y_0\,\mathbf{j}$.
- **Polar relations:** $r = \sqrt{x^2 + y_0^2}$ and $\tan\theta = y_0/x$; equivalently $\cos\theta = x/r$ and $\sin\theta = y_0/r$.
- **Polar basis.** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x/r & -y_0/r & 0 \\ y_0/r & x/r & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \frac{x^2 + y_0^2}{r^2} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- The inverse relation $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$ is the one used in 3.4.

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the motion along x is uniformly accelerated, so $x(t)$ and $v(t)$ follow by integrating $\ddot{x} = a_x$ with the initial conditions $\dot{x}(0) = v_0$ and $x(0) = 0$ (Barrow's rule):

$$\dot{x}(t) = v_0 + \int_0^t a_x\,dt' = v_0 + [a_x t']_0^t = v_0 + a_x t - a_x \cdot 0 \equiv v(t), \qquad x(t) = 0 + \int_0^t (v_0 + a_x t')\,dt' = \left[v_0 t' + \frac{1}{2}a_x t'^2\right]_0^t = \left(v_0 t + \frac{1}{2}a_x t^2\right) - 0$$

- The polar position follows from the geometric relations of Phase 2: $\mathbf{r} = r\,\mathbf{e}_r$ with $r(t) = \sqrt{\left(v_0 t + \frac{1}{2}a_x t^2\right)^2 + y_0^2}$ and $\theta(t) = \operatorname{arccot}\left(\dfrac{v_0 t + \frac{1}{2}a_x t^2}{y_0}\right)$, with the branch $\operatorname{arccot} \in (0, \pi)$ (so $\theta(0) = \pi/2$, and θ stays in $(0, \pi)$ even if $a_x < 0$ makes x negative). The velocity

and acceleration come from differentiating these relations.

(a) Proof that $\dot{r} = \dfrac{v(t)\,x(t)}{r(t)}$

> **Pedagogical Justification:** differentiate implicitly the relation $r^2 = x^2 + y_0^2$ with respect to time.

$$2r\dot{r} = 2x\dot{x} + 0 = 2x\,v(t) \implies \dot{r} = \frac{v(t)\,x(t)}{r(t)} \quad \blacksquare$$

(b) Proof that $\dot{\theta} = -\dfrac{v(t)\,y_0}{r(t)^2}$

> **Pedagogical Justification:** use $\cot\theta = x/y_0$, which is valid for every t (including $x = 0$, where $\tan\theta$ is infinite), and differentiate with the chain rule: $\frac{d}{dt}\cot\theta = -\csc^2\theta\,\dot{\theta}$ and $\csc^2\theta = 1/\sin^2\theta = r^2/y_0^2$.

$$-\csc^2\theta\,\dot{\theta} = \frac{\dot{x}}{y_0} = \frac{v(t)}{y_0} \implies -\frac{r^2}{y_0^2}\dot{\theta} = \frac{v}{y_0} \implies \dot{\theta} = -\frac{v(t)\,y_0}{r(t)^2} \quad \blacksquare$$

(c) Proof that $\ddot{r} = \dfrac{v^2(t)\,y_0^2 + a_x\,x(t)\,r^2(t)}{r(t)^3}$

> **Pedagogical Justification:** differentiate $\dot{r} = vx/r$ with the quotient rule $\left(\frac{u}{w}\right)' = \frac{\dot{u}w - u\dot{w}}{w^2}$, with $u = vx$ and $w = r$.

$$\dot{u} = \dot{v}x + v\dot{x} = a_x x + v^2, \qquad \dot{w} = \dot{r} = \frac{vx}{r}, \qquad \ddot{r} = \frac{\left(a_x x + v^2\right) r - vx\,\dfrac{vx}{r}}{r^2} = \frac{\left(a_x x + v^2\right) r^2 - v^2 x^2}{r^3}$$

$$\left(a_x x + v^2\right) r^2 - v^2 x^2 = a_x x r^2 + v^2\left(r^2 - x^2\right) = a_x x r^2 + v^2 y_0^2 \implies \ddot{r} = \frac{v^2 y_0^2 + a_x\,x\,r^2}{r^3} \quad \blacksquare$$

(d) Proof that $\ddot{\theta} = y_0 \dfrac{2v^2(t)\,x(t) - a_x\,r^2(t)}{r(t)^4}$

> **Pedagogical Justification:** differentiate $\dot{\theta} = -y_0\,v\,r^{-2}$ with the quotient rule, using $\dot{v} = a_x$ and $\frac{d}{dt}\left(r^2\right) = 2r\dot{r} = 2vx$:

$$\ddot{\theta} = -y_0\,\frac{\dot{v}r^2 - v\,\frac{d}{dt}(r^2)}{r^4} = -y_0\,\frac{a_x r^2 - 2v^2 x}{r^4} = y_0\,\frac{2v^2x - a_x r^2}{r^4} \quad \blacksquare$$

Position, velocity and acceleration in polar coordinates

- **Position and velocity:** $\mathbf{r} = r\,\mathbf{e}_r$ and, with (a) and (b),

$$\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta = v\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right)$$

- **Acceleration** $\mathbf{a} = a_r\mathbf{e}_r + a_\theta\mathbf{e}_\theta$ from the polar formula:

$$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{v^2y_0^2 + a_x x r^2}{r^3} - r\left(\frac{vy_0}{r^2}\right)^2 = \frac{a_x x r^2}{r^3} = a_x\frac{x}{r}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{y_0\left(2v^2x - a_x r^2\right)}{r^3} - \frac{2v^2xy_0}{r^3} = -a_x\frac{y_0}{r} \implies \mathbf{a} = a_x\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right)$$

3.2 Forces

Not applicable: kinematics-only problem. The motion law along x is given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The acceleration a_x is prescribed.

3.4 Integration and Answers

Repeating the problem in Cartesian coordinates

> **Pedagogical Justification:** in the fixed Cartesian basis the unit vectors are constant, so the description is immediate; the polar result must agree with it through the change of basis.

$$\mathbf{r} = x(t)\,\mathbf{i} + y_0\,\mathbf{j}, \qquad \mathbf{v} = v(t)\,\mathbf{i}, \qquad \mathbf{a} = a_x\,\mathbf{i}$$

- Converting with $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta = \frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta$:

$$\mathbf{v} = v\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right), \qquad \mathbf{a} = a_x\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right)$$

These equal the polar results of 3.1 identically.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[\dot r] = [vx/r] = \mathrm{m/s}$ and $[\dot\theta] = [vy_0/r^2] = \mathrm{s^{-1}}$; $[\ddot r] = \dfrac{(\mathrm{m/s})^2\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s^2}$ and $[\ddot\theta] = \dfrac{\mathrm{m}\cdot(\mathrm{m/s})^2\cdot\mathrm{m}}{\mathrm{m}^4} = \mathrm{s^{-2}}$.
- **Limit $x \to \infty$ (aircraft far away):** $r \to x$, so $\dot r \to v$, $\dot\theta \to 0$, $a_r \to a_x$ and $a_\theta \to 0$: the line of sight aligns with the flight direction and the motion is purely radial.
- **Aircraft overhead ($x = 0, t = 0$):** $\dot r = 0$, $\dot\theta = -v_0/y_0$, $a_r = 0$ and $a_\theta = -a_x$; with $\theta = \pi/2$, $\mathbf{a} = a_x\mathbf{i} = -a_x\mathbf{e}_\theta$.
- **Invariance of the modulus:** $a_r^2 + a_\theta^2 = a_x^2\,\dfrac{x^2 + y_0^2}{r^2} = a_x^2$, equal to $\|a_x\mathbf{i}\|^2$ because $[{}_0R_1]$ is orthogonal.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 4 in its solution keys, but the statement gives the four formulas (a) to (d) as targets, and they are proved above. Sympy confirms the four identities and $a_r = a_x x/r$, $a_\theta = -a_x y_0/r$ with zero residual.

Source: Problems.pdf

PROBLEM 05

Plane and Glider

Original statement · Problems.pdf, p. 7

Aerospace Towed Flight

An airplane A is flying horizontally with a constant speed v_A, and is towing a glider B. The glider is gaining altitude. The tow cable has a length r, and the angle θ formed between the tow cable and the horizontal is increasing at a constant rate of $\dot{\theta}$. We want to determine the magnitude of the velocity, v_B^0, and the acceleration, a_B^0, of the glider as a function of θ and its derivatives.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a tow plane A flies horizontally at constant speed v_A and tows a glider B that gains altitude. The tow cable has length r and forms an angle θ with the horizontal.
- **Hypotheses:** v_A and the cable length r are constant; $\mathbf{a}_A = \mathbf{0}$. The statement says that $\dot{\theta}$ is constant; the answer is requested as a function of θ and its derivatives, so we keep $\theta(t)$ general and particularise at the end ($\ddot{\theta} = 0$).
- **Constraints and degrees of freedom:** we place the origin at the altitude of A, so $y_A = 0$. The taut cable of constant length keeps B at the distance r from A: $(x_B - x_{A0} - v_A t)^2 + y_B^2 = r^2$. This is a holonomic constraint that depends explicitly on time (rheonomic) because the position of A is prescribed, and it is bilateral while the cable is taut. B then has $\mathrm{CDOF} = 2 - 1 = 1$: the angle θ.
- **Coordinates:** $\theta(t)$ for the glider relative to A; the position of A along the flight direction is $x_A(t) = x_{A0} + v_A t$.
- **Parameters and units:** v_A [m/s], r [m], θ [rad], $\dot{\theta}$ [rad/s], $\ddot{\theta}$ [rad/s^2].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$ with $\mathbf{i}_0$ horizontal in the direction of flight of A and $\mathbf{j}_0$ vertical, upward.
- **Relative position of B with respect to A:** the glider is behind and above the plane, so the vector from A to B points at the angle θ above the negative horizontal. Its components in $\mathcal{B}_0$ follow from the geometry of the cable:

$$\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

This is the only reading compatible with the glider gaining altitude as θ increases ($r\sin\theta$ grows with θ for $0 < \theta < \pi/2$).

PHASE 3

Derivation

3.1 Kinematics

Position of the glider

> **Pedagogical Justification:** the position of B is the position of A plus the relative position, all written in the fixed basis, so that no derivative of a unit vector is needed.

$$\mathbf{r}_A = (x_{A0} + v_A t)\,\mathbf{i}_0, \qquad \mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = (x_{A0} + v_A t - r\cos\theta)\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

Velocity of the glider

> **Pedagogical Justification:** differentiate $\mathbf{r}_B$ in $\mathcal{S}_0$ with the chain rule, since θ depends on time and r is constant:

$$\frac{d}{dt}(-r\cos\theta) = r\sin\theta\,\dot\theta, \qquad \frac{d}{dt}(r\sin\theta) = r\cos\theta\,\dot\theta$$

$$\mathbf{v}_B^0 = \left(v_A + r\dot\theta\sin\theta\right)\mathbf{i}_0 + r\dot\theta\cos\theta\,\mathbf{j}_0, \qquad v_B^0 = \sqrt{\left(v_A + r\dot\theta\sin\theta\right)^2 + \left(r\dot\theta\cos\theta\right)^2}$$

Acceleration of the glider

Pedagogical Justification: differentiate each component of $\mathbf{v}_B^0$ with the product and chain rules:

$$\frac{d}{dt}\left(v_A + r\dot\theta\sin\theta\right) = r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta, \qquad \frac{d}{dt}\left(r\dot\theta\cos\theta\right) = r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta$$

$$\mathbf{a}_B^0 = \left(r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta\right)\mathbf{i}_0 + \left(r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta\right)\mathbf{j}_0$$

- **Magnitude.** Expanding both squares,

$$\left(r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta\right)^2 = r^2\ddot\theta^2\sin^2\theta + 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta + r^2\dot\theta^4\cos^2\theta$$

$$\left(r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta\right)^2 = r^2\ddot\theta^2\cos^2\theta - 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta + r^2\dot\theta^4\sin^2\theta$$

and adding them, the cross terms $\pm 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta$ cancel:

$$\left(a_B^0\right)^2 = r^2\ddot\theta^2\left(\sin^2\theta + \cos^2\theta\right) + r^2\dot\theta^4\left(\cos^2\theta + \sin^2\theta\right) = r^2\ddot\theta^2 + r^2\dot\theta^4$$

3.2 Forces

Not applicable: kinematics-only problem. The motion of A and the cable geometry are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. No dynamic unknown appears.

3.4 Integration and Answers

Pedagogical Justification: the answers requested are the magnitudes of the velocity and of the acceleration of B in S_0 as functions of θ and its derivatives:

$$\boxed{v_B^0 = \sqrt{\left(v_A + r\dot{\theta}\sin\theta\right)^2 + \left(r\dot{\theta}\cos\theta\right)^2}}, \qquad \boxed{a_B^0 = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}}$$

- **Expanded speed.** Expanding the square in v_B^0 (valid for any $\theta(t)$, using $\sin^2\theta + \cos^2\theta = 1$): $v_B^0 = \sqrt{v_A^2 + 2v_A r\dot{\theta}\sin\theta + r^2\dot{\theta}^2}$.
- **Constant rate of the statement.** With $\dot{\theta}$ constant, $\ddot{\theta} = 0$ and $a_B^0 = r\dot{\theta}^2$.

PHASE 4

Interpretation & Checks

- **Decoupling from the cruise speed:** the acceleration does not depend on v_A, because $\mathbf{a}_A = \mathbf{0}$ (uniform rectilinear translation). The glider accelerates only through its circular motion relative to A.
- **Limiting case $\ddot{\theta} = 0$:** the acceleration is purely centripetal, $a_B^0 = r\dot{\theta}^2$, directed along the cable toward A (circular motion of B around A).
- **Limit $\dot{\theta} \equiv 0$ (identically, so $\ddot{\theta} = 0$ too):** $\mathbf{v}_B^0 = v_A\,\mathbf{i}_0$ and $\mathbf{a}_B^0 = \mathbf{0}$: the glider moves rigidly with the plane. (If $\dot{\theta}$ only vanishes at one instant, $a_B^0 = r|\ddot{\theta}|$ there, not zero.)
- **Dimensional check:** $[r\ddot{\theta}] = [r\dot{\theta}^2] = \mathrm{m}\cdot\mathrm{s}^{-2}$ and $[v_A + r\dot{\theta}\sin\theta] = \mathrm{m/s}$, so a_B^0 is in $\mathrm{m/s^2}$ and v_B^0 in m/s.

- **Check against the official key:** the key of `Problems.pdf` is $v_B^0 = \sqrt{(v_A + r\dot\theta\sin\theta)^2 + (r\dot\theta\cos\theta)^2}$ and $a_B^0 = \sqrt{r^2\ddot\theta^2 + r^2\dot\theta^4}$ (read on the rendered page), identical to the result above. Sympy confirms both squares for an arbitrary function $\theta(t)$ with zero residual.

Source: Problems.pdf

PROBLEM 06

Particle on Sinusoidal Curve

Original statement · Problems.pdf, p. 7

Frenet-Serret Frame

A point particle P travels along a curve given by:

$$y = \sin x$$

with a velocity along x given by $\dot{x} = 1$.

(a) Find the intrinsic, unitary vector basis associated to the curve.

(b) Express x as a function of an arc length parameter, s. You can leave your solution as a function of an integral.

(c) Find the tangent and normal acceleration components of P as a function of x.

(d) Write the radius of curvature of the trajectory as a function of x.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P travels along the curve $y = \sin x$ with horizontal velocity $\dot{x} = 1$.
- **Hypotheses:** the data are in SI units, so $y = \sin x$ means $y = (1\ \mathrm{m})\sin(x/1\ \mathrm{m})$ and $\dot{x} = 1\ \mathrm{m/s}$. We take $x(0) = 0$, so $x(t) = t$ (in these units).
- **Constraints and degrees of freedom:** the curve $y = \sin x$ is a holonomic, scleronomic constraint in the plane, so $\mathrm{CDOF} = 2 - 1 = 1$. The coordinate is x, whose motion law is given.
- **Coordinates:** x (and the arc length s along the curve).

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with the curve in the Oxy plane.
- **Parametric curve:** $\mathbf{r}(x) = x\,\mathbf{i}_0 + \sin x\,\mathbf{j}_0$, with $\dfrac{d\mathbf{r}}{dx} = \mathbf{i}_0 + \cos x\,\mathbf{j}_0$. The differential arc length is

$$ds = \left\| \frac{d\mathbf{r}}{dx} \right\| dx = \sqrt{1 + \cos^2 x}\, dx \equiv L(x)\, dx$$

- **Intrinsic basis.** $\mathbf{e}_t$ is the unit tangent and $\mathbf{e}_n$ points toward the centre of curvature (Notes Sec. 2.4). Both are obtained in 3.1. In the plane the normal can be written as the left normal $\mathbf{k}_0 \times \mathbf{e}_t$ up to a sign, because the curve has inflection points where the centre of curvature switches sides:

$$\mathbf{k}_0 \times \mathbf{e}_t = \frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{L(x)}, \qquad \mathbf{e}_n = \operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L(x)} \quad (\sin x \neq 0)$$

- **Change of basis.** The matrix with the components of $\mathbf{e}_t$, $\mathbf{e}_n$ and $\mathbf{e}_b$ in $\mathcal{B}_0$ is, for $\sin x \neq 0$,

$$[{}_0R_{\text{tnb}}] = \frac{1}{L}\begin{pmatrix} 1 & \operatorname{sgn}(\sin x)\cos x & 0 \\ \cos x & -\operatorname{sgn}(\sin x) & 0 \\ 0 & 0 & -L\operatorname{sgn}(\sin x) \end{pmatrix}, \qquad \det = \frac{1}{L^3}\left(-L\operatorname{sgn}(\sin x)\right)\left(-\operatorname{sgn}(\sin x) - \operatorname{sgn}(\sin x)\cos^2 x\right) = 1, \qquad RR^T = I$$

For $\sin x > 0$, $\mathbf{e}_n$ is the right normal $-\mathbf{k}_0 \times \mathbf{e}_t$ and $\mathbf{e}_b = -\mathbf{k}_0$; for $\sin x < 0$ it is the left normal and $\mathbf{e}_b = \mathbf{k}_0$. In both cases the triad is right-handed.

PHASE 3

Derivation

3.1 Kinematics

Velocity and acceleration

Pedagogical Justification: the position is expressed in the fixed basis, so the derivatives of the unit vectors vanish. With $\dot{x} = 1$ and $\ddot{x} = 0$ (chain rule $\dot{y} = \cos x\,\dot{x}$):

$$\mathbf{v} = \dot{x}\,\mathbf{i}_0 + \cos x\,\dot{x}\,\mathbf{j}_0 = \mathbf{i}_0 + \cos x\,\mathbf{j}_0, \qquad v(x) = \sqrt{1+\cos^2 x} = L(x)$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{x}\,\mathbf{i}_0 + \left(-\sin x\,\dot{x}^2 + \cos x\,\ddot{x}\right)\mathbf{j}_0 = 0\,\mathbf{i}_0 - \sin x\,\mathbf{j}_0 = -\sin x\,\mathbf{j}_0$$

(a) Intrinsic unit basis

- **Tangent:** $\mathbf{e}_t = \dfrac{d\mathbf{r}/dx}{\|d\mathbf{r}/dx\|} = \dfrac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{L}$.
- **Pedagogical Justification (normal):** by the definition of the course, $d\mathbf{e}_t/ds = \mathbf{e}_n/\rho$, so $\mathbf{e}_n$ is the direction of $d\mathbf{e}_t/ds$. We differentiate with respect to x (product rule, with $\frac{d}{dx}L^{-1} = -\frac{1}{2}L^{-3}\left(-2\cos x\sin x\right) = \sin x\cos x\,L^{-3}$) and use $ds = L\,dx$:

$$\frac{d\mathbf{e}_t}{dx} = \frac{\sin x\cos x}{L^3}\left(\mathbf{i}_0 + \cos x\,\mathbf{j}_0\right) + \frac{1}{L}\left(-\sin x\,\mathbf{j}_0\right) = \frac{\sin x\cos x\,\mathbf{i}_0 + \left[\sin x\cos^2 x - \sin x\left(1+\cos^2 x\right)\right]\mathbf{j}_0}{L^3} = -\frac{\sin x}{L^3}\left(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0\right)$$

$$\frac{d\mathbf{e}_t}{ds} = \frac{1}{L}\frac{d\mathbf{e}_t}{dx} = -\frac{\sin x}{L^4}\left(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0\right)$$

- The norm of this vector is the curvature, and its direction is $\mathbf{e}_n$. The vector $-\cos x\,\mathbf{i}_0 + \mathbf{j}_0$ has norm L, so

$$\left\|\frac{d\mathbf{e}_t}{ds}\right\| = \frac{|\sin x|}{L^3} = \frac{1}{\rho}, \qquad \mathbf{e}_n = -\operatorname{sgn}(\sin x)\,\frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{L} = \operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L}$$

- For $0 < x < \pi$ ($\sin x > 0$) the arch is concave down and $\mathbf{e}_n$ points downward; for $\pi < x < 2\pi$ it points upward. At the inflection points $x = k\pi$ the curvature vanishes and $\mathbf{e}_n$ is not defined ($\rho \to \infty$).
- **Binormal:** $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = -\operatorname{sgn}(\sin x)\,\mathbf{k}_0$, constant in each arch.

(c) Tangent and normal acceleration components

> **Pedagogical Justification:** project the acceleration on the intrinsic vectors:

$$a_t = \mathbf{a}\cdot\mathbf{e}_t = (-\sin x\,\mathbf{j}_0)\cdot\frac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{L} = -\frac{\sin x\cos x}{\sqrt{1+\cos^2 x}}$$

$$a_n = \mathbf{a}\cdot\mathbf{e}_n = (-\sin x\,\mathbf{j}_0)\cdot\operatorname{sgn}(\sin x)\,\frac{\cos x\,\mathbf{i}_0 - \mathbf{j}_0}{L} = \operatorname{sgn}(\sin x)\,\frac{\sin x}{L} = \frac{|\sin x|}{\sqrt{1+\cos^2 x}} \geq 0$$

- **Check:** $a_t^2 + a_n^2 = \dfrac{\sin^2 x\left(\cos^2 x + 1\right)}{1+\cos^2 x} = \sin^2 x = \|\mathbf{a}\|^2$.

3.2 Forces

Not applicable: kinematics-only problem. The curve and the motion law $\dot{x} = 1$ are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by the statement.

3.4 Integration and Answers

(b) Arc-length parameter $s(x)$

> **Pedagogical Justification:** integrate the metric differential from $x = 0$ (taking $s(0) = 0$):

$$s(x) = \int_0^x \sqrt{1 + \cos^2 u}\, du$$

The integral cannot be written with elementary functions (extension: it is an incomplete elliptic integral of the second kind); the statement allows leaving it as an integral. Since the integrand satisfies $\sqrt{1 + \cos^2 u} \geq 1 > 0$, $s(x)$ is strictly increasing, so it can be inverted to give $x(s)$.

(d) Radius of curvature

> **Pedagogical Justification:** from $a_n = v^2/\rho$, with $v^2 = 1 + \cos^2 x$:

$$\rho(x) = \frac{v^2}{a_n} = \frac{1 + \cos^2 x}{|\sin x|/\sqrt{1 + \cos^2 x}} = \boxed{\frac{\left(1 + \cos^2 x\right)^{3/2}}{|\sin x|}}$$

As a check (extension), this is the standard curvature formula of a plane graph, $\rho = \dfrac{\left(1 + y'^2\right)^{3/2}}{|y''|}$, for $y' = \cos x$ and $y'' = -\sin x$.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[v] = \mathrm{m/s}$, $[a_t] = [a_n] = \mathrm{m/s^2}$, $[\rho] = [v^2/a_n] = \mathrm{m}$ and $[s] = \left[\int \sqrt{1 + \cos^2 u}\, du\right] = \mathrm{m}$ (using the length scale 1 m inside the sine).

- **Special points:** at the inflection points $x = k\pi$, $a_n = 0$ and $\rho \to \infty$ (locally straight). At the crest $x = \pi/2$: $v = 1$ m/s, $a_t = 0$, $a_n = 1$ m/s^2 and $\rho = 1$ m, equal to $1/|y''| = 1$ m; there $\mathbf{e}_n = -\mathbf{j}_0$, toward the centre of curvature below the crest. At $x = \pi/4$: $\rho = (3/2)^{3/2}/(\sqrt{2}/2) = 3\sqrt{3}/2 = 2.598$ m.
- **Arc length:** $s(\pi) = \int_0^{\pi} \sqrt{1 + \cos^2 u}\, du = 3.8202$ m, greater than the horizontal distance π m, as it must be.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 6 in its solution keys, so the results are verified by independent derivation: sympy reproduces a_t, a_n, $a_t^2 + a_n^2 = \sin^2 x$, the curvature $\|d\mathbf{e}_t/ds\| = |\sin x|/L^3$ and $\rho(\pi/2) = 1$, $\rho(\pi/4) = 3\sqrt{3}/2$; numerical quadrature gives $s(\pi) = 3.82020$ m.

Source: Problems.pdf